

PRODUCT SOURCE OF TRUTH

Portfolio Intelligence / News Explainer

An evidence-first system for investigating unusual portfolio and security-level movements.

Canonical product brief

Version 1.0 | 03 October 2026 | Product definition

PURPOSE

This document is the canonical source of truth for product, architecture, implementation and evaluation decisions. Future GitHub work should trace back to these principles, boundaries and phased outcomes.

CORE POSITION

Accelerate investigation. Preserve evidence. Communicate uncertainty. Keep human judgment final.

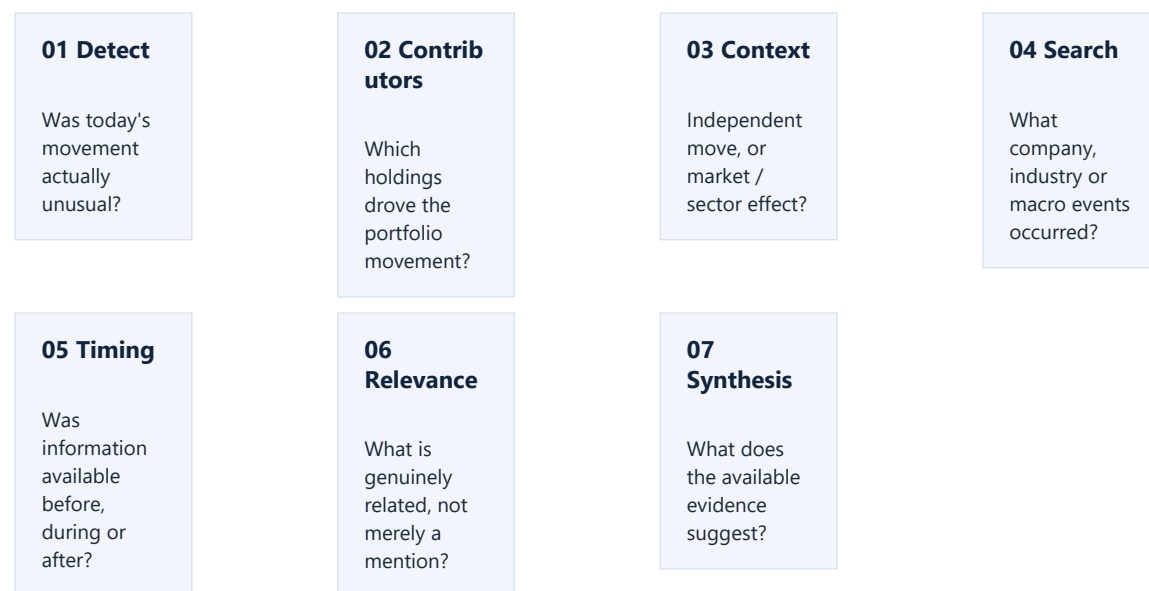
01 / PRODUCT VISION

Portfolio Intelligence is an AI-assisted investigation system that helps investors and financial professionals understand potential drivers of unusual portfolio and security-level movements.

When a portfolio experiences a significant move, the system identifies which holdings contributed most, determines whether those securities behaved abnormally relative to broader market or sector conditions, retrieves time-relevant financial news and evidence, and produces a grounded explanation with citations and uncertainty.

The system is designed to accelerate investigation, not determine causality or replace human financial judgment.

02 / PROBLEM



This workflow is fragmented across market-data platforms, news sources, search, portfolio analytics and manual reasoning. The result can be a slow and inconsistent investigation process, especially when multiple holdings move simultaneously.

03 / TARGET USERS

PRIMARY / Investment & portfolio analyst

Professionals monitoring portfolios who need to investigate unexpected performance quickly: investment analysts, portfolio analysts, wealth-management professionals, financial advisors and risk analysts.

SECONDARY / Sophisticated individual investor

Investors seeking evidence about potential drivers rather than generic financial-news summaries. The MVP remains designed primarily around the analyst investigation workflow.

04 / JOB TO BE DONE

“When my portfolio or one of its holdings moves unexpectedly, help me quickly identify what drove the portfolio movement and surface credible, time-relevant evidence that may explain the security movement so that I can investigate it efficiently.”

Not the job: “Tell me definitively why the stock moved.” That would imply causality the system usually cannot establish.

05 / CURRENT WORKFLOW & PAIN-POINT HYPOTHESES

Portfolio moves → analyst notices movement → identifies largest contributors → checks security price chart → compares against index / sector → searches financial news → opens multiple articles → checks publication timestamps → evaluates relevance → forms an explanation → communicates findings.

Fragmentation

Information exists across multiple systems.

Information overload

Many articles mention a company without explaining the movement.

Timing ambiguity

Relevant reporting may appear after the price movement.

Context loss

A decline may simply reflect a sector or market decline.

Repetition

Analysts repeat similar steps for every movement.

LLM trust

Generic summaries can sound convincing despite weak evidence.

These are hypotheses to validate through user interviews, not established facts.

06 / PRODUCT HYPOTHESIS

If we automatically combine portfolio attribution, abnormal-movement detection, market context and temporally aligned evidence retrieval with a grounded AI explanation layer, users can investigate unusual portfolio movements faster while retaining visibility into the underlying evidence and uncertainty.

The system uses AI to accelerate analysis, not replace the analyst.

07 / PROPOSED PRODUCT WORKFLOW

STAGE	SYSTEM ACTION	QUESTION ANSWERED
1. Monitor	Calculate portfolio performance and security-level contribution.	What happened to the portfolio?
2. Identify	Rank holdings by contribution to portfolio movement.	Which securities matter?
3. Detect	Evaluate movement against history using volatility, z-score, thresholds, anomaly detection or intraday windows.	Is the movement unusual?
4. Contextualize	Compare the security with market, sector and relevant peers.	Company-specific or broader?
5. Evidence	Create an event, retrieve and rank temporally aligned sources.	What credible evidence exists?
6. Explain	Synthesize cited evidence, confidence and alternatives.	What may have coincided, and what remains unknown?

Example attribution

Holding	Return	Portfolio contribution
NVDA	-5.2%	-1.10%
MSFT	-2.1%	-0.61%
JPM	+0.7%	+0.08%

08 / EVENT STRUCTURE

Monitoring and context stages produce a structured investigation event. This event becomes the input to the evidence pipeline.

FIELD	EXAMPLE
Security	NVDA
Portfolio contribution	-1.10%
Security return	-5.2%
Abnormal return	-3.8%
Movement window	10:34 AM - 10:51 AM
Market context	S&P; 500 -0.4%
Sector context	Semiconductors -1.1%
Investigation required	YES

09 / EVIDENCE RETRIEVAL

Retrieve information around the detected movement from financial news APIs, company announcements, SEC filings, earnings information, and market / sector news.

PRE-MOVEMENT Available before the detected movement. Strongest temporal basis for a contemporaneous hypothesis.	DURING MOVEMENT Published or released inside the movement window. Requires precise timestamp handling.	POST-MOVEMENT Potentially explanatory reporting, but not evidence that market participants had it beforehand.
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Publication timing affects what conclusions can reasonably be drawn.

10 / EVIDENCE RANKING

Retrieved information is evaluated before it reaches the LLM. The final evidence score combines signals rather than relying exclusively on embedding similarity.

SIGNAL	ROLE	CAUTION
Semantic relevance	Embeddings and similarity assess proximity to the company / event.	Similarity alone is insufficient.
Temporal relevance	Measures proximity to the movement window.	Distinguish event time from publication time.
Financial relevance	Detects meaningful events: earnings, guidance, regulation, management, products, analyst actions, litigation and macro events.	Mentions are not necessarily material.
Sentiment	FinBERT or another finance-language model adds a directional signal.	Does not establish relevance or causality.
Source quality	Gives more consideration to reliable and original sources.	Coverage gaps and bias remain.

11 / AI EXPLANATION

The LLM receives selected evidence, not unrestricted internet content. Every material explanation points to supporting evidence.

- What potential factors coincided with this movement?
- Was the movement company-specific or broader?
- What evidence was available before the movement?
- What appeared afterward?

12 / CONFIDENCE & ABSTENTION

No explanation is better than an unsupported explanation.

If evidence is insufficient, conflicting or temporally inappropriate, the system says so.

EVIDENCE CONFIDENCE / LOW

Available sources do not provide sufficient evidence to identify a credible company-specific driver for this movement. Broader semiconductor weakness may have contributed, but the available evidence does not support a stronger conclusion.

13 / CONVERSATIONAL INVESTIGATION

Later versions provide an investigation assistant capable of follow-up questions:

- Why did NVDA contribute so much to my portfolio loss?
- Was the movement company-specific?
- Show only evidence published before the decline.
- What happened in the semiconductor sector?
- What evidence contradicts this explanation?

The system should not be designed only to confirm its original answer.

14 / AI VS DETERMINISTIC DESIGN

LAYER	USE FOR
Deterministic / statistical	Portfolio returns and attribution; timestamps; market-data calculations; movement detection; data validation; filtering; basic financial calculations.
ML / NLP	Financial sentiment; semantic similarity; anomaly detection where appropriate; evidence classification and ranking.
LLM	Evidence synthesis; natural-language explanation; conversational investigation.

Architectural rule: Do not use an LLM for something ordinary code can calculate reliably. This is AI-first thinking without AI-everywhere thinking.

15 / HUMAN IN THE LOOP

The system assists rather than silently making investment conclusions. Users retain access to original evidence, timestamps, market context, confidence and alternative explanations, and ultimately decide how to interpret the event.

The system does not provide investment recommendations.

16 / MINIMUM VIABLE PRODUCT

Input: A weighted sample portfolio containing NVDA, MSFT, AAPL, AMZN and JPM.

#	MVP CAPABILITY
1	Retrieve market data
2	Calculate portfolio returns
3	Calculate holding contribution
4	Detect significant security movement
5	Compare with market / sector context
6	Retrieve relevant news
7	Validate timestamps
8	Rank evidence
9	Generate grounded explanation
10	Display sources
11	Abstain when evidence is insufficient

This is sufficient for a serious first release.

17 / NON-GOALS

The MVP will not attempt to:

- Predict stock prices
- Recommend buying or selling securities
- Execute trades
- Prove news causality
- Replace financial analysts
- Provide personalized investment advice
- Monitor every financial information source
- Build proprietary foundation models
- Provide institutional-grade portfolio accounting

These boundaries keep the project technically realistic and intellectually defensible.

18 / SUCCESS METRICS

DIMENSION	METRICS
Product	Investigation time; evidence usefulness; user confidence and understanding.
Retrieval	Precision@K; recall where labels allow; temporal validity rate; evidence relevance.
AI quality	Citation correctness; faithfulness; unsupported-claim rate; explanation completeness; abstention quality.
System	Latency; API failure rate; data completeness; retry success; cost per investigation.

The system is measured, not merely declared to “work.” These metrics also establish the dashboard, reporting and KPI layer needed for product governance.

19 / RESPONSIBLE AI PRINCIPLES

Grounding Material claims require supporting evidence.	Transparency Users can inspect the evidence.	Uncertainty Confidence must be communicated.
Abstention Weak evidence should produce no strong conclusion.	Temporal integrity Evidence timing must be considered explicitly.	Human control AI supports investigation rather than making investment decisions.

20 / MAJOR RISKS

RISK	WHY IT MATTERS	PRODUCT RESPONSE
False causality	Coincidence does not establish causation.	Use association language; show context and alternatives.
Hallucination	An LLM may introduce unsupported explanations.	Constrain to selected evidence; verify citations; abstain.
Publication delay	Reporting may appear after the underlying event.	Separate event time from publication time.
Data quality	Market or news timestamps may be inconsistent.	Validate, normalize and expose provenance.
Retrieval bias	News APIs may miss relevant information.	Communicate coverage; use multiple evidence types later.
Source reliability	Sources vary substantially in quality.	Rank original and reliable sources higher.
Market complexity	Movements often have multiple interacting causes.	Support multiple hypotheses and contradictory evidence.
Evaluation difficulty	Definitive ground truth may not exist.	Use layered metrics and expert review.

These are not hidden weaknesses. They are part of the case study because acknowledging them demonstrates critical thinking about AI in financial systems.

21 / FIVE PRODUCT PRINCIPLES

01	Evidence before explanation.
02	Deterministic where possible; AI where useful.
03	Association is not causation.
04	Abstain rather than fabricate certainty.
05	Human judgment remains final.

22 / INITIAL ROADMAP

PHASE	FOCUS	OUTCOME
MVP	Detect & Explain	Portfolio attribution → movement detection → news retrieval → grounded explanation.
V1	Context	Market + sector + peer comparison.
V2	Trust	Evaluation framework → confidence → abstention → failure analysis.
V3	Investigation	Interactive dashboard + conversational investigation.
V4	Intelligence	Better ranking → multiple evidence types → contradictory evidence → richer financial context.
V5	Enterprise readiness	Monitoring → audit trail → permissions → governance → operating model.

IMPLEMENTATION HANDOFF

Product brief □ → create GitHub repository → establish project structure → add README + source of truth → make initial commit → begin Phase 1.

This order preserves a clean development history from product definition through architecture, implementation and evaluation. All material scope decisions should be checked against this source of truth.